

1. Administrative & Study Identification

Full Study Title: A phase 3, randomized, double-blind, study to assess efficacy and safety of ianalumab (VAY736) versus placebo in warm autoimmune hemolytic anemia (wAIHA) patients who failed at least one line of treatment **Short Title/Acronym:** VAYHIA
Protocol Number: CVAY736012301 **NCT Number:** NCT05648968 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate that either dose of ianalumab induces a durable hemoglobin (Hb) response compared to placebo in patients with warm autoimmune hemolytic anemia (wAIHA) who have failed at least one line of treatment. **Secondary Objectives:** To demonstrate that either dose of ianalumab maintains a durable hemoglobin response that is sustained beyond the end of the treatment period, compared to placebo. Additional objectives include assessing the duration of response, time to response, complete response rate, need for rescue treatment, and overall safety and tolerability.

2.2 Background & Rationale To address the significant unmet medical need for patients with wAIHA, a rare autoimmune disorder characterized by autoantibody destruction of red blood cells. Current standard-of-care options (such as corticosteroids and off-label rituximab) are associated with toxicities and rarely achieve long-term, durable responses after treatment completion. Ianalumab targets the BAFF receptor (BAFF-R) and has a unique dual mechanism of action that depletes autoreactive B cells, providing a complementary physiological approach to potentially restore self-tolerance and maintain safe hemoglobin levels.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (with an optional open-label crossover period for placebo patients not responding) **Randomization:** Randomized, Parallel Assignment (1:1:1 ratio to low-dose ianalumab, high-dose ianalumab, or placebo)

3.2 Planned Enrollment & Duration Target **NS\$:** Approximately 90 adult participants **Number of Sites:** Multicenter **Estimated Study Start:** 13-Dec-2022 **Estimated Primary Completion:** Q4-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients ≥ 18 years of age at the time of signing consent. 2. Documented primary or secondary wAIHA by a positive direct antiglobulin test specific for anti-IgG or anti-IgA. 3. Insufficient response to, or relapsed after at least one line of treatment, including patients with steroid resistance, dependence, or intolerance. 4. Hemoglobin concentration at screening and at Week 1 ≥ 5 g/dL and < 10 g/dL, associated with the presence of symptoms related to anemia. 5. Dose of supportive care must be stable for at least 4 weeks prior to randomization.

4.2 Exclusion Criteria 1. Other forms of autoimmune hemolytic anemia or unrelated hematologic diseases. 2. Active or uncontrolled infections. 3. Prior or ongoing use of certain prohibited B-cell depleting therapies.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Low-dose ianalumab, high-dose ianalumab, or matching placebo administered once every 4 weeks. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Initial 25-week treatment period, with participants who do not achieve durable response on placebo having the option to cross over to receive open-label high-dose ianalumab.

5.2 Concomitant & Prohibited Therapy Permitted: Specific stable supportive care medications for wAIHA as defined in the protocol. Rescue medications and RBC transfusions are permitted if clinically indicated to ensure patient safety. **Prohibited:** Initiation of new unapproved systemic immunosuppressants or targeted B-cell therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 4 weeks	Informed Consent, Medical History, Hb Labs, Direct Antiglobulin Test
2	Baseline	Day 0
3	Randomization, First IV Infusion	Treatment & Efficacy
4	Weeks 1-25	Bi-weekly (every other week) onsite visits: IV Dosing (Q4W), Hemoglobin assessments, Safety Labs
5	Follow-up	End of Study
6	Week 25 to 2 Years	Monthly safety visits for the first 20 weeks post-treatment, then quarterly up to 2 years. Responders are followed for efficacy up to 39 months post-randomization.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Durable hemoglobin (Hb) response, defined as Hb ≥ 10 g/dL and an increase of ≥ 2 g/dL from baseline for ≥ 8 consecutive weeks between Week 9 and Week 25, in the absence of rescue medication or prohibited treatment. **Measurement Method:** Central/Local Laboratory Analysis.

7.2 Secondary & Exploratory Endpoints

- 1. Duration of durable Hb response:** Time from the first Hb assessment showing durable response to the earliest of: Hb < 10 g/dL for ≥ 2 consecutive weeks, start of any rescue/prohibited treatment, or death.
- 2. Complete Response Rate:** Normalization of hemoglobin levels (≥ 11 g/dL for women, ≥ 12 g/dL for men) and no evidence of hemolysis without the need for red blood cell transfusions.
- 3. Time to response and Quality of response** across treatment groups.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting during all bi-weekly, monthly, and quarterly follow-up visits. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocols for severe adverse reactions. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms implemented by Novartis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all tolerability analyses. **Sample Size Justification:** Appropriately powered ($N=90$) to detect statistically significant improvements in durable hemoglobin response compared to placebo. **Handling of Missing Data:** Rescue medication administration or early discontinuation of the investigational product will result in the participant being categorized as a non-responder.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized onsite and remote clinical monitoring. **Audit Trail:** Robust eCRF systems are utilized to track dosing, lab inputs, and protocol deviations.

11. Study Identifiers & Governance

Primary ID: CVAY736012301 **Registry Number:** NCT05648968 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** VAYHIA **Official Regulatory Title:** A phase 3, randomized, double-blind, study to assess efficacy and safety of ianalumab (VAY736) versus placebo in warm autoimmune hemolytic anemia (wAIHA) patients who failed at least one line of treatment

12. Clinical Framework (wAIHA Specific)

Primary Condition: Warm Autoimmune Hemolytic Anemia (wAIHA) **Diagnosis Threshold:** Baseline Hemoglobin ≥ 5 g/dL and < 10 g/dL **Medical Methodology:** B-cell depletion and BAFF-R signaling blockade **Optimization Target:** Durable hemoglobin response ≥ 10 g/dL and an increase of ≥ 2 g/dL from baseline

13. Study Procedures & Methodology

Management Pathway: Double-blind, placebo-controlled intravenous dosing with optional open-label crossover for non-responders **Education Protocol (HCP):** Protocol and safety mechanism training **Education Protocol (Patient):** Informed consent, reporting mechanisms, and study design orientation **Quality Audit Mechanism:** Central and site-level trial monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to 39 months post-randomization for durable responders **Onsite Visit Cadence:** Bi-weekly during the treatment and primary endpoint follow-up period **Remote/Long-term Monitoring Frequency:** Monthly for the first 20 weeks after the last dose, then quarterly for up to 2 years **Checklist Review Interval:** Per protocol visit schedule

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				